

Project: End-to-End SOC Detection & Incident Response Portfolio

Mission

Build a production-quality cybersecurity portfolio project that demonstrates the complete workflow of a modern Security Operations Center (SOC), from log ingestion and threat detection to investigation, incident response, and reporting.

This project is not a collection of isolated cybersecurity exercises. It must function as a coherent security operations environment where every alert, detection rule, dashboard visualization, and incident report is connected and tells a clear investigative story.

The final result should convince a recruiter, hiring manager, or security team that the creator understands how real-world blue teams operate and can contribute to SOC, Threat Detection, Detection Engineering, or Incident Response roles.

Primary Objective

Create an end-to-end SOC lab portfolio containing:

1. A live SIEM dashboard accessible through a public link.
2. A GitHub repository containing professionally written Sigma detection rules.
3. Multiple incident investigation reports generated from alerts produced by those Sigma rules.
4. Architecture documentation explaining the environment.
5. MITRE ATT&CK mapping throughout the project.

Every component must be interconnected.

A recruiter should be able to:

Dashboard → Alert → Sigma Rule → Investigation Report → Evidence → Response Actions

without encountering disconnected artifacts.

Quality Standard

The project must resemble work produced by a junior SOC analyst, detection engineer, or blue-team practitioner rather than a student completing tutorial exercises.

Avoid placeholder content.

Avoid generic screenshots without context.

Avoid theoretical explanations that are not supported by evidence generated in the lab.

Every alert, dashboard panel, and incident report must be backed by actual telemetry collected within the environment.

Technical Environment

Build a realistic security monitoring lab using technologies such as:

- ELK Stack (Elasticsearch, Logstash, Kibana)
- Splunk
- Wazuh
- Security Onion
- Sysmon
- Windows Event Logs
- Linux Audit Logs

Generate realistic security events and attacks for analysis.

Examples:

- PowerShell abuse
 - Brute-force authentication attempts
 - Credential dumping
 - Suspicious process execution
 - Privilege escalation activity
 - Persistence mechanisms
 - Lateral movement simulations
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Dashboard Requirements

The SIEM dashboard must provide operational visibility.

Include:

- Security alert overview
- Authentication activity
- Failed login monitoring
- Process creation monitoring
- Endpoint activity
- Severity-based alert tracking
- MITRE ATT&CK coverage visualization

- Incident trends

Dashboard screenshots alone are insufficient.

Provide:

- Live dashboard access
 - Documentation explaining each panel
 - Data sources
 - Detection logic references
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Sigma Detection Engineering Requirements

Create a professional Sigma rule repository.

Each rule must include:

- Clear title
- Description
- Detection logic
- Log source definition
- MITRE ATT&CK mapping
- False positive considerations
- Severity level
- Testing methodology

Example categories:

- PowerShell encoded commands
- Credential dumping
- Suspicious parent-child process relationships
- Brute-force attacks
- Persistence mechanisms
- Lateral movement indicators

Each Sigma rule must correspond to evidence that appears in the SIEM dashboard and triggers a documented investigation.

Incident Response Requirements

Create at least three complete incident investigations.

Each report must contain:

Executive Summary

Explain what occurred and why it matters.

Detection

Describe the alert that initiated the investigation.

Investigation Timeline

Chronological analysis of events.

Evidence

Include:

- Logs
- Screenshots
- Indicators of compromise
- Process details
- Network activity

MITRE ATT&CK Mapping

Identify attacker techniques and tactics.

Root Cause Analysis

Explain how and why the activity occurred.

Containment Actions

Document remediation steps.

Lessons Learned

Discuss detection improvements and future recommendations.

Example Investigation Cases

Case 1: PowerShell Encoded Command Execution

MITRE: T1059.001

Case 2: Brute Force Authentication Attempt

MITRE: T1110

Case 3: Credential Dumping Activity

MITRE: T1003

Each case must produce:

- Dashboard evidence
- Sigma detection
- Alert generation
- Investigation report

Documentation Requirements

Create a professional README that includes:

- Project overview
- Architecture diagram
- Lab setup instructions
- Data sources
- Detection methodology
- MITRE ATT&CK coverage
- Screenshots
- Public dashboard link
- GitHub repository structure

The README should immediately communicate that this is an operational SOC portfolio rather than a classroom exercise.

Success Criteria

A recruiter reviewing the project for five minutes should conclude:

- The candidate understands SIEM operations.
- The candidate can write detections.
- The candidate can investigate alerts.
- The candidate can document incidents.
- The candidate understands MITRE ATT&CK.
- The candidate can communicate technical findings professionally.
- The candidate possesses practical blue-team skills applicable to a real SOC environment.

The final portfolio should be credible enough to be showcased during interviews and used as primary evidence of hands-on cybersecurity experience.